LOOK INSIDE

PKG-005 · Energy & Power Pack

Sample pages from: Off Grid Electrical Solar Schematic

These are unaltered pages from the actual guide, taken from the current fixed library (Stage 4, 2026-10-30). Every EOTWSG guide opens with an action card, states its safety boundaries up front, and ends with printable fields for your household.

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Off-Grid Electrical & Solar Power Schematic

EOTWSG | MODULE 01 | REV 2026-09-22 | STATUS: SOURCE-VERIFIED ONLY

Authored by: EOTWSG

REVISION LOG

Date	Revision	Description
2026-09-22	REV 2026-09-22	Front-matter standardization: cover block, revision log, action card, and local fields added; source normalized to house Typst dialect.

ACTION CARD

Use this guide to nEC/NFPA 70 and CEC Part I requirements must be checked against the adopted local editions and amendments.Project review basis: NEC 2023 / CEC 2024 as specified in the EOTWSG technical baseline.. Carry one copy in the master box and one copy in the working binder.

Step	Do this now	Completion check	Stop / escalate when
1	Confirm this copy is current: check the module ID (01) and REV date on the cover.	Module ID and REV noted on the working copy.	If the REV predates 2026-09-22, replace this copy before use.
2	Work through the Design Basis section against your household's actual people, conditions, and equipment.	Section completed with household-specific entries; gaps flagged.	If a required capability is missing, flag it in the binder and resolve before the next drill.
3	Record completion, date, and any deviations in the working binder; schedule the next review.	Binder entry dated and filed.	If the household situation changes, re-run the guide.

PURPOSE AND ACTIVATION

Purpose: NEC/NFPA 70 and CEC Part I requirements must be checked against the adopted local editions and amendments.Project review basis: NEC 2023 / CEC 2024 as specified in the EOTWSG technical baseline.

Activation triggers: utility failure, evacuation order, sudden service uncertainty, or a planned verification drill. This module is usable on its own; linked module IDs preserve the shared decision order from the master index.

6.0 kW PV — 48 V LiFePO4 — NEC 2023 / CEC 2024

Design Basis

Parameter	Locked input	Review note
PV array	6.0 kW roof-mounted	Verify roof attachment, wind uplift, fire access, and module/string layout.
Battery	48 V DC nominal LiFePO4	Confirm BMS, thermal limits, enclosure, ventilation, and listing.
Conductors	Marine-grade UL 1426 tinned copper	Minimum 10 AWG PV string runs; 4/0 AWG battery interconnects.
Codes	NEC 2023 / CEC 2024	Local amendments and AHJ review govern.

Single-Line System Overview

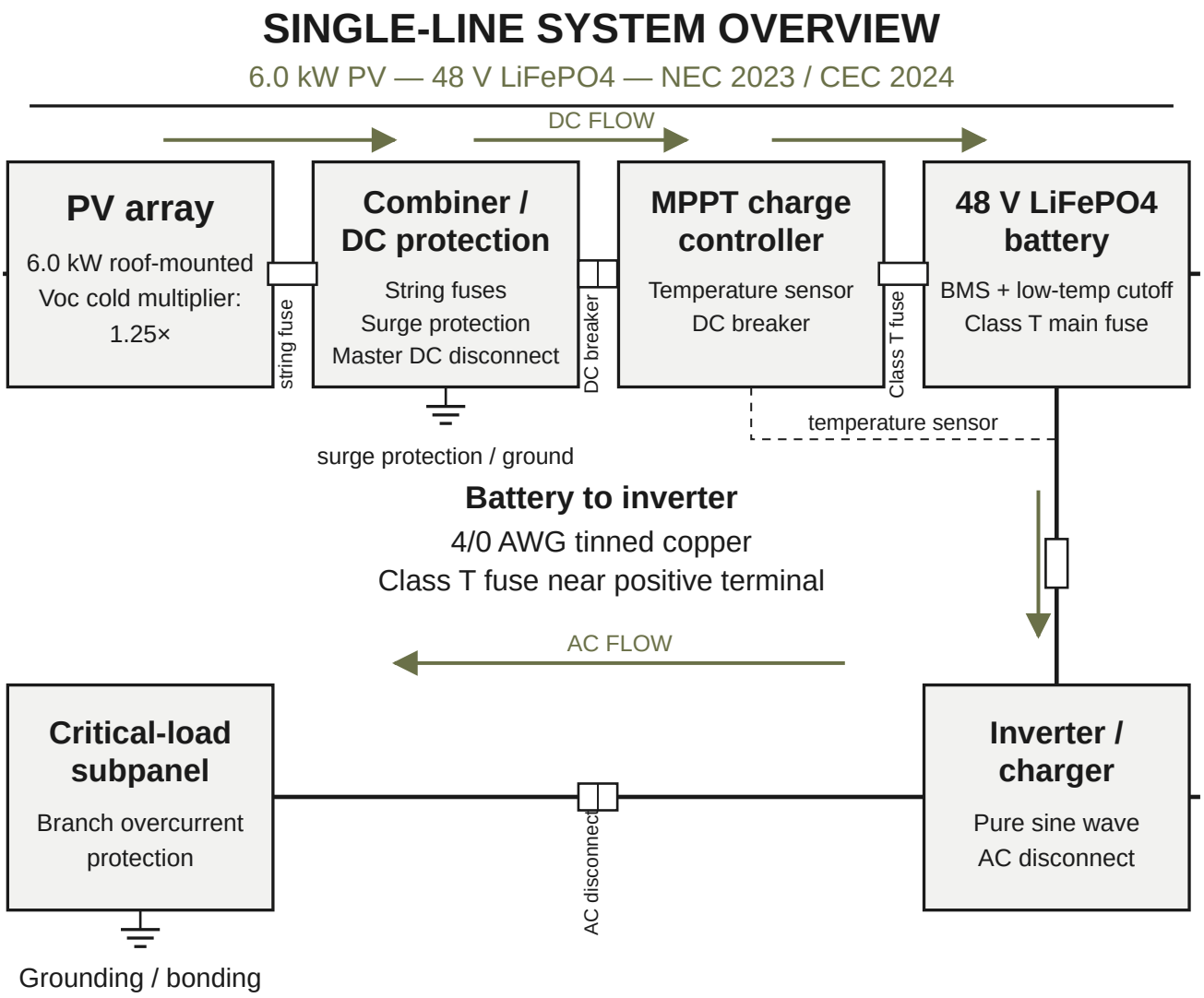


Figure 1: Single-line system overview.

Cold-Weather Voltage and Overcurrent Rules

1. Calculate the cold-weather array open-circuit voltage as 1.25 × the manufacturer's maximum string Voc before selecting the controller, conductors, disconnects, and enclosures.
2. Size breakers and fuses at 125% of the continuous load rating unless the governing code and equipment listing require a more restrictive rule.
3. Coordinate PV string fuses, combiner protection, DC disconnects, surge protection, grounding, and rapid-shutdown requirements with the AHJ.
4. Place the main positive battery cable Class T fuse as close to the battery positive terminal as the manufacturer and code permit; do not omit the fuse.

Bill of Materials / Review Schedule

System	Minimum / target	Required feature	Engineer / installer confirmation

PV strings / modules	6.0 kW total	String Voc, Isc, 1.25× cold Voc, 10 AWG minimum PV wire, MC4 or listed equivalent	**Confirm:** _____
Combiner	As designed	Touch-safe string fuses, surge arrestor, grounding lug, weather-rated enclosure	**Confirm:** _____
DC conduit run	As designed	6 AWG copper or engineered equivalent; Schedule 80 where required	**Confirm:** _____
MPPT controller	48 V battery output	Input voltage above cold-weather Voc; DC breaker; sensor	**Confirm:** _____
Battery bank	48 V LiFePO4	BMS, cell monitoring, low-temp charge cutoff, listed enclosure	**Confirm:** _____
Battery interconnect	4/0 AWG minimum	Tinned copper, protected routing, Class T fuse on positive	**Confirm:** _____
Inverter / charger	As engineered	Pure sine wave, AC overcurrent, disconnect, grounding/bonding	**Confirm:** _____

ELECTRICAL SAFETY

DC arcs and battery fault currents can cause fire, burns, explosion, or death. De-energize, lock out, verify absence of voltage, use listed equipment, and use qualified personnel. This sheet does not establish final conductor ampacity, fault-current ratings, or code compliance.

References

NEC/NFPA 70 and CEC Part I requirements must be checked against the adopted local editions and amendments. Project review basis: NEC 2023 / CEC 2024 as specified in the EOTWSG technical baseline.

Credit: Excerpts, schematics, and tactical protocols adapted from the End of the World Survival Guide (EOTWSG) by Matthew Danielson. Reproduce only for off-grid resilience and survival training.

SAFETY BOUNDARIES

WARNING - MODULE LIMITS

Do not perform medical, electrical, or structural procedures beyond this guide's stated limits. When in doubt, stop and consult qualified professionals. This guide is a preparedness aid, not a guarantee of safety. Follow current instructions from local emergency management and authorities.

- This module directs specific actions; those actions carry the safety limits stated above. Do not improvise beyond them.
- Use nonviolent de-escalation and withdrawal as the primary security response.
- If conditions exceed the household's capability, escalate to professional responders.

LOCAL FIELDS AND RECORDS

Use these printable fields and keep them at the front of the working binder:

Field	**Fill-in**
Household / location name	**Name:** _____
Primary contact	**Name and number:** _____
Date completed	**Date:** _____
Next review date	**Date:** _____